

Chapter 4 Continuous Time Markov Chains

4.1 Definition and Examples

4.2 Computing the Transition Probability

$\{X_t, t \geq 0\}$ is a **continuous time Markov chain**:

finite or countably infinite state space;

$$P(X_{s+t}=j|X_s=i, X_{s_k}=i_k, 0 \leq k \leq n) = P(X_t=j|X_0=i),$$

$\forall 0 \leq s_0 < \dots < s_n < s$ and possible $i_0, \dots, i_n, i, j$.

Remark

- Given present, future is independent of past.
- The transition is temporally homogeneous.
- No (1-step prob.) transition matrix: no minimal steps.

Example 4.1

Y_n : a discrete time MC with trans. prob. $u(i, j)$;

$N(t)$: a Poisson process with rate λ . (ind. of Y .)

Then, $X_t = Y_{N(t)}$ is a continuous time Markov chain.

(X_t jumps according to $u(i, j)$ at arrivals of $N(t)$.)

Remark

- Deterministic time $\rightarrow$ exponential time.
- lack of memory of exp. distribution $\Rightarrow$ Markovian.

Transition Probabilities & Transition Rates

Transition probability (for each $t > 0$):

$$p_t(i, j) = P(X_t = j | X_0 = i).$$

In **Example 4.1** Transition probability of $X_t = Y_{N(t)}$:

$$p_t(i, j) = \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} u^n(i, j).$$

Theorem (Chapman-Kolmogorov Equation)

$$\sum_k p_s(i, k) p_t(k, j) = p_{s+t}(i, j) .$$

Jump rate from i to j

$$q(i, j) = \lim_{h \rightarrow 0} \frac{p_h(i, j)}{h} \quad \text{for } j \neq i .$$

In **Example 4.1** $q(i, j) = \lambda u(i, j)$ is the jump rate: the chain leaves i at rate λ and goes to j with prob. $u(i, j)$.

Transition rate matrix Q

$$Q(i, j) = \begin{cases} q(i, j) & \text{if } j \neq i , \\ -\sum_{j \neq i} q(i, j) = -\lambda_i & \text{if } j = i . \end{cases}$$

off-diagonal ≥ 0 ; row sum $= 0$.

Transition rates $\rightarrow$ Transition probabilities

Kolmogorov's backward/forward equation:

$$p'_t = Qp_t / p'_t = p_t Q, \quad p_0 = I. \quad (\Leftarrow \text{C-K eq.})$$

Example 4.7 (Rate λ Poisson Process)

$$p_t(i, j) = e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!} \quad \text{for } j \geq i; \quad 0 \quad \text{otherwise.}$$

$$Q = \begin{bmatrix} -\lambda & \lambda & 0 & \dots \\ 0 & -\lambda & \lambda & \dots \\ 0 & 0 & -\lambda & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Check Kolmogorov's backward/forward equation.

A closed form solution: $p_t = e^{Qt} = \sum_{n=0}^{\infty} Q^n \frac{t^n}{n!}$, $t \geq 0$.

For explicit trans. prob.: solving system of diff. eqs.

Example 4.8 (Two-state Chain) $Q = \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}$.

The backward equation is

$$\begin{bmatrix} p'_t(1,1) & p'_t(1,2) \\ p'_t(2,1) & p'_t(2,2) \end{bmatrix} = \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix} \cdot \begin{bmatrix} p_t(1,1) & p_t(1,2) \\ p_t(2,1) & p_t(2,2) \end{bmatrix}$$

As a result, $p_t(1,1) = \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t}$,
 $p_t(2,1) = \frac{\mu}{\lambda+\mu} - \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t}$,
 $p_t(1,2) = 1 - p_t(1,1)$, $p_t(2,2) = 1 - p_t(2,1)$.

Example 4.2 (Poisson Process) A Poisson process $X(t)$ with rate λ : CTMC with transition rate

$$q(n, n+1) = \lambda \text{ for all } n \geq 0.$$

Transition Rate $\longrightarrow$ CTMC

- 1st description: when at state i ,
assign independent $\exp(q(i,j))$ alarms for each $j \neq i$;
jump to j corresponding to the first alarm.
(Can show that $p_h(i,j) = q(i,j)h + o(h)$.)
- 2nd description: when at state i ,
wait for $\exp(\lambda_i)$ time ($\lambda_i = \sum_{j \neq i} q(i,j) \in (0, \infty)$);
jump to $j (\neq i)$ with probability $r(i,j) = q(i,j)/\lambda_i$.
(Easy to simulate CTMC following these steps.)

4.3 Limiting Behavior

- **NO periodicity issue**

MC X_t irreducible: i to j in finite jumps, $\forall i, j$. i.e.,
 $\exists i = k_0, k_1, \dots, k_n = j$ s.t. $q(k_{m-1}, k_m) > 0, 1 \leq m \leq n$.
($\Rightarrow p_t(i, j) > 0, \forall i, j, t$.)

- **Stationary Distribution:** $\pi p_t = \pi, \forall t > 0$.
 $\iff \pi Q = 0$.

$$(\pi Q = 0 \iff \sum_{k \neq j} \pi_k q(k, j) = \lambda_j \pi_j = \sum_{k \neq j} \pi_j q(j, k):$$

rate in = rate out, balanced.)

- **Detailed Balance Condition:** $\exists \pi$, s.t.

$$\pi(k)q(k, j) = \pi(j)q(j, k) \text{ for all } j \neq k .$$

($\implies \pi$ is a stationary distribution. Not vice versa.)

Example 4.12 (Birth-Death Chain Satisfies DBC)

Suppose that $S = \{0, 1, \dots, N\}$ with $N \leq \infty$ and

$$q(n, n+1) = \lambda_n \text{ for } n < N ,$$

$$q(n, n-1) = \mu_n \text{ for } n > 0 .$$

λ_n, μ_n : the birth, death rate when n individuals.

$$\pi(n) = \frac{\lambda_{n-1} \cdot \lambda_{n-2} \dots \lambda_1 \cdot \lambda_0}{\mu_n \cdot \mu_{n-1} \dots \mu_2 \cdot \mu_1} \pi(0) .$$

Theorem

X_t : CTMC, irreducible, $\exists$ a stationary distr. π . Then

$$\lim_{t \rightarrow \infty} p_t(i, j) = \pi(j).$$

Sojourn Time, Return Time, Stationary Distr.:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t 1_{X_s=j} ds = \frac{1}{\lambda_j \bar{E}_j \bar{T}_j}$$

$$\pi(j) = \lim_{t \rightarrow \infty} \frac{1}{t} E \left(\int_0^t 1_{X_s=j} ds \right) = \frac{1}{\lambda_j \bar{E}_j \bar{T}_j}.$$

Example 4.10 (LA Weather Chain)

Sunny (1) $\xrightarrow[\text{mean}=3]{\text{exp. distr. \# days}}$ Smoggy (2) $\xrightarrow[\text{mean}=4]{\text{exp. distr. \# days}}$

Rainy (3) $\xrightarrow[\text{mean}=1]{\text{exp. distr. \# days}}$ Sunny (1)

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & 0 \\ 0 & -\frac{1}{4} & \frac{1}{4} \\ 1 & 0 & -1 \end{bmatrix} \end{matrix}; \quad \pi(1) = \frac{3}{8}, \pi(2) = \frac{4}{8}, \pi(3) = \frac{1}{8}. \\ \text{(Not DBC)}$$

Limiting fraction of sunny, smoggy, rainy days: $\frac{3}{8}, \frac{4}{8}, \frac{1}{8}$.